Restoring Fish Passage in the Fraser Region - 2025

Executive Summary

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Prepared for the Habitat Conservation Trust Foundation and the Ministry of Transportation and Infrastructure

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on behalf of the Society for Ecosystem Restoration in Northern British Columbia

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Claude (Anthropic) assisted with literature synthesis, drafting, and technical writing. All scientific interpretation, data analysis, and conclusions are the responsibility of the authors.

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[Executive Summary (PDF)]

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_fraser_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from fish_passage_fraser_2025_reporting.pdf.

Since 2023, the Society for Ecosystem Restoration Northern British Columbia (SERNbc) has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Nechako River, Lower Chilako River, Morkill River, Upper Fraser River, and François Lake watershed groups, which are sub-basins of the Upper Fraser River watershed. That work was supported internally by SERNbc and through the Ministry of Transportation and Infrastructure, and produced the region's 2023 reporting. The project is now also supported by the Habitat Conservation Trust Foundation, alongside Skeena fish passage restoration planning, together forming the Northern British Columbia Fish Passage Restoration project (CAT26-0-636).

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document 2025 field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time.

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Fish passage assessment procedures conducted through SERNbc in the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups since 2023 are amalgamated online within the Assessment Data Summary appendix of the report found here which includes links to project reporting for each site.

In 2025, the project area was expanded to include the Willow river, Lower Salmon River, and Tabor River watershed groups. Fish passage assessments were completed at 12 new sites, focusing on structures with potential barriers to upstream fish movement using standard provincial criteria.

Habitat confirmation assessments were conducted at 6 sites within the Tabor River and Willow River watershed groups, covering approximately 4.3 km of stream length. Detailed habitat metrics were documented alongside eDNA samples to evaluate habitat quality and presence of fish species.

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 33 samples were collected across 15 streams, alongside 5 field and office blanks run as protocol controls.

Monitoring was conducted at two sites near Prince George, including post-remediation evaluations at Tabor Creek (PSCIS crossing 196085) and Bittner Creek (PSCIS crossing 196200). Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

The components of this initiative are designed to complement one another, building the state of knowledge regarding watershed health and directing restoration effort to where it will do the most good. Passage assessments and the provincial model establish where fish can reach; habitat confirmations, fish sampling and environmental DNA establish where they are; floodplain delineation and aerial imagery establish how infrastructure has altered the valley bottom; climate departure and water temperature establish the trajectory; effectiveness monitoring tests whether the work delivered. Together they make explicit what is known, what is not, and what should be measured next. Making that evidence usable is part of the work — data collected on structured digital forms, methods documented alongside the code that implements them, layers published to open catalogues, and reporting written so partners, First Nations, regulators and tenure holders can question it and build on it — so understanding accumulates across the program rather than being rebuilt each season.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices, building trust, and staying committed to a longer-term transformation.

A single definitive ranking of remediation priorities is not achievable, because the criteria that determine whether a crossing is worth remediating are not measured in common units, and the deciding constraint sits outside the project in the capacity and willingness of infrastructure owners and tenure holders to support implementation across multi-year timelines. What can be done systematically is to maintain a well-characterised set of candidate sites, so that when a tenure holder is in a position to act, the ecological value of the opportunity and the appropriate scope of work are already understood.

Looking ahead, priorities for the program are:

- Continue and extend the long-term effectiveness monitoring program so decisions elsewhere are better informed.
- Continue environmental DNA sampling as an annual, region-wide layer, building single-year detections into distribution and trend information.
- Extend functional floodplain mapping and the climate departure analysis across the project area, so watershed condition sits alongside barrier status wherever prioritization is done.
- Commission engineering designs at the Ministry of Transportation and Infrastructure crossings identified in previous reporting, and advance the highest ranked sites assessed this year.
- Resolve species use through targeted fish sampling where habitat ranks high but detections are absent or unresolved.
- Continue working with First Nations, stewardship groups, educational institutions and tenure holders, feeding monitoring results back into prioritization and design as adaptive management.